

Laboratory Exercise 12 (Exam Simulation)

Programming Exercise

Develop a Python program to allow a police station to keep track of any contacts that may have occurred among the guests of a suspicious hotel.

The hotel has a text file, named `customers.txt`, which contains, in each row, the information of a different hotel guest. Each row contains **4 fields separated by commas**. The first field is the guest's name, the second is the mobile phone number, and the remaining two fields are the check-in and check-out days of the guest at the hotel.

For simplicity, the check-in and check-out days are represented by integers (between 1 and 366).

The program must allow the police station to **search for any "contacts" of suspicious guests**. For each suspicious guest, the program must display the **name and phone number of all other hotel guests who stayed at the hotel for at least one day together with that guest**. The list of contacts must be displayed in alphabetical order, by the guest's name.

The "suspicious" guests for whom contacts need to be searched are stored, one per row, in the file `suspects.txt`.

For each line of this file, the search must be performed, and the list described above must be printed. The case where the suspicious guest had no contact with other guests must be distinguished by displaying the text as shown in the following example.

Given the following `customers.txt` file:

```
Mario Rossi,3471234567,100,110
Paolo Verdi,3353334444,105,112
Maria Azzurri,3398887777,98,104
Anna Neri,06989855,95,100
Guido Guidi,3331112221,90,93
```

Assuming the `suspects.txt` file has the following content:

```
Anna Neri
Paolo Verdi
Guido Guidi
```

The program should produce the following output:

```
** Contacts of the guest: Anna Neri: **
    Contact with Maria Azzurri, phone 3398887777
    Contact with Mario Rossi, phone 3471234567
** Contacts of the guest: Paolo Verdi: **
    Contact with Mario Rossi, phone 3471234567
** Contacts of the guest: Guido Guidi: **
    The guest Guido Guidi had no contacts
```

Open-ended Question 1

What is the result of the following sum, in 8-bit two's complement representation: 10111110 + 01110011?

- A. Overflow
- B. 100110001
- C. 00110001

(Answer: C)

Open-ended Question 2

Given a 4 MB memory with a data bus of 2 bytes, how many address lines are required to address the memory?

- A. 21
- B. 22
- C. 24

(Answer: A: $2^{(22)/2}$)

Open-ended Question 3

Which of the following pieces of information about a group of students can be stored in a Python set? Justify your answer.

- A. For each student, their last name
- B. For each student, the list of their grades
- C. For each student, the tuple (student ID, average exam score)

(Answer: C. A is not suitable because last names can be duplicated, B is not suitable because sets cannot contain mutable elements, such as lists.)